# Sprint Report #1

## Team Members

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## Project Customer

Innovative Systems L.L.C.

## Customer Description

Innovative Systems is a developer and provider of software and hardware for the independent telecommunications industry. More than 1200 systems are deployed at locations across the United States and Canada.

## Goal of Project

The goal of this project is to design a single game server that communicates with a STB display mechanism. Also this project will provide a tool for future STB game developers to deploy games in a simple standardized way.

## Customer Needs

- Single Game Server:
    - Communication Protocol
    - Game Logic
    - Game Resources
    - Game AI
    - Supporting 100-1000 users
- STB Display Mechanism
    - Modular
    - Dynamic at Runtime
- Virtual APMAX
    - Communication Protocol
        - STB -> Communication Layer 1 -> APMAX
        - APMAX -> Communication Layer 1 -> STB
    - Secure Communication
- Unit Testing

# Overview of the Project

The overall goal of this project is to perform a proof of concept of a multi-player game environment for the Set Top Boxes in the in-place APMax environment. A framework will have to be set up on both the STBs and a central server to host gameplay.

# Project Environment

## Project Boundaries

The games that run on the STBs must be small and very efficient. STBs have very little memory available, and games have to be re-downloaded each time it is played. Therefore, all player info and statistics must be stored on the server, and then retrieved at run-time. All games will be created in a custom C++/QT environment. The server process will be written in C# and be able to be run on a Windows/Azure server.

## Project Context

This project is a continuation off of the 2011/2012 Set Top Box Game project, which was also hosted by Innovative Systems. The reason for wanting multi-player games is simply that playing games with other people is more fun than being limited to playing alone. This will keep customers interested in the game feature and will make Innovative Systems product more marketable.

## Technical Environment

Our technical environment will exist on a Virtual Box Image of Ubuntu provided by the STB box developers in Mitchell. The Game Server environment will be designed on a Windows Platform using Visual Studio 2012 and according server software that will be determined in future sprints.

### Current Systems Overview

- The games will run on Amino Set Top Boxes. These STBs run a custom build of QT, binded to C++. The game server software will be written in C# and must be able to run under .NET 4.5. The server should be able to operate in a Windows or Azure operating system.
- The multi-player game environment will consist of:
    - One or more APMax units
    - Amino Set Top Boxes
    - One game server
    - One or more individual servers dedicated to A.I. processing

The multi-player game environment will be spread cross country and distributed over many systems already in place. The environment follows the client-server model with an added element in the middle, the APMax, known as middleware.

# Project Deliverables

- Client Contract
- Mission Statement

- Requirement Document
- Design Document
- CONOPS Document

## Backlog Outline

- CmdLink Design and Integration
- Server Plugin Interface
- Research: Server operating systems
- Research: Security for the server
- Research: UDP vs TCP
- Research QML
- Server Tests
- Server Logic
- Server-Game Pluggin Interface
- Server Communication
- Prove out and implement QUiLoader
- Generic-ize each STB game's code to work with Model/View architecture
- Build Template STB GUI
- Implement Virtual APMax

## Future Issues or Problems

- STB Runtime design using QtUILoader (can’t test until v9 is stable)
- Set Top Box memory constraints
- STB Version Updates